Czech Technical University in Prague

Faculty of Electrical Engineering
Department of Cybernetics
Multi-robot Systems

My Thesis Title Can Span Multiple Lines

Bachelor's Thesis

John Smith

Prague, January 2021

Study programme: Electrical Engineering and Information Technology
Branch of study: Artificial Intelligence and Biocybernetics

Supervisor: Ing. My Supervisor, Ph.D.

Acknowledgments

Firstly, I would like to express my gratitude to my supervisor.

Swap this for the Thesis Assignment, when you have it from the department.

Declaration

I declare that presented work was developed independently, and that I have listed all sources of information used within, in accordance with the Methodical instructions for observing ethical principles in preparation of university theses.

Date ....

Abstract

The study of autonomous Unmanned Aerial Vehicles (UAVs) has become a prominent sub-field of mobile robotics.

Keywords Unmanned Aerial Vehicles, Automatic Control

Abstrakt

Výzkum na poli autonomních bezpilotních prostředků (UAV) se stal významným oborem mobilní robotiky.

Klíčová slova Bezpilotní Prostředky, Automatické Řízení

Abbreviations

GPS Global Positioning System

LiDAR Light Detection and Ranging

UAV Unmanned Aerial Vehicle

Contents

1 Introduction **1**
1.1 Related works . . . 1
1.2 Contributions . . . 1
1.3 Mathematical notation . . . 1

2 Extraction of Topological Information from Binary Masks **2**
2.1 Problem Formulation . . . 2
2.2 Preliminaries . . . 2
2.3 Binary Mask Creation . . . 2
2.4 Euclidean Distance Transform . . . 2
2.5 Binary Mask Smoothing . . . 2
2.6 Binary Mask Skeletonization . . . 3
2.7 Binary Mask Contours . . . 3

3 Curve Representation of Extracted Topology **4**
3.1 Problem Formulation . . . 4
3.2 Preliminaries . . . 4
3.3 Endpoint and Junction Pixels Detection . . . 4
3.4 Polyline Creation Algorithm . . . 4
3.4.1 Polyline from Endpoint to Junction or Endpoint . . . 4
3.4.2 Polyline from Junction to Junction . . . 5

4 How to write thesis in LaTeX **6**
4.1 Versioning with git . . . 6
4.2 Forming paragraphs . . . 6
4.3 Linguistic anti patterns . . . 6
4.3.1 Narrative . . . 6
4.3.2 Pronouns . . . 6
4.4 Mathematical notation with LaTeX . . . 6
4.4.1 Common errors . . . 7
4.4.2 Equations . . . 7
4.5 Using footnotes . . . 8
4.6 Referencing document elements . . . 8
4.7 Abbreviations with Acronym . . . 8
4.8 Units of measurements with Siunitx . . . 8
4.9 Hyphens and dashes . . . 8
4.10 Double quotation marks . . . 9
4.11 2D Diagrams with Tikz . . . 9
4.12 Data plots with PGFPlots . . . 9

4.13 3D Plots with Sketch 10
4.14 Image collages with Subfig 10
4.15 Citations with Biblatex 10
4.16 Image overlays with Tikz 11
4.17 General tips 11

5 Conclusion **13**

6 References **14**

A Appendix A **15**

1 Introduction

First, introduce the reader to the research topic. Start with the most general view and slowly converge to the particular field, sub-field, and the challenges you face. You can cite others' work here [1].

1.1 Related works

This section should contain related state-of-the-art works and their relation to the author's work. We usually cite the original works like this [3]. You can also cite multiple papers at once like this [1], [2].

1.2 Contributions

This section should describe the author's contributions to the field of research.

1.3 Mathematical notation

It is a good practice to define basic mathematical notation in the introduction. See Table 1.1 for an example.

$\mathbf{x}$, $\boldsymbol{\alpha}$	vector, pseudo-vector, or tuple
$\hat{\mathbf{x}}$, $\hat{\boldsymbol{\omega}}$	unit vector or unit pseudo-vector
$\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$	elements of the *standard basis*
$\mathbf{X}, \boldsymbol{\Omega}$	matrix
$\mathbf{I}$	identity matrix
$x = \mathbf{a}^\intercal \mathbf{b}$	inner product of $\mathbf{a}$, $\mathbf{b} \in \mathbb{R}^3$
$\mathbf{x} = \mathbf{a} \times \mathbf{b}$	cross product of $\mathbf{a}$, $\mathbf{b} \in \mathbb{R}^3$
$\mathbf{x} = \mathbf{a} \circ \mathbf{b}$	element-wise product of $\mathbf{a}$, $\mathbf{b} \in \mathbb{R}^3$
$\mathbf{x}_{(n)} = \mathbf{x}^\intercal \hat{\mathbf{e}}_n$	n^{th} vector element (row), $\mathbf{x}, \mathbf{e} \in \mathbb{R}^3$
$\mathbf{X}_{(a,b)}$	matrix element, (row, column)
x_d	x_d is *desired*, a reference
$\dot{x}, \ddot{x}, \dddot{x}, \ddddot{x}$	1st, 2nd, 3rd, and 4th time derivative of x
$x_{[n]}$	x at the sample n
$\mathbf{A}, \mathbf{B}, \mathbf{x}$	LTI system matrix, input matrix and input vector
SO(3)	3D special orthogonal group of rotations
SE(3)	*SO(3)* $\times \mathbb{R}^3$, special Euclidean group

Table 1.1: Mathematical notation, nomenclature and notable symbols.

2 Extraction of Topological Information from Binary Masks

In order to correctly create features of high fidelity environments such as roads, rivers and biomes, it is essential to extract the necessary information from a segmentation image. In this chapter, we present the key concepts related to topological information in binary masks and introduce our proposed solutions.

2.1 Problem Formulation

2.2 Preliminaries

2.3 Binary Mask Creation

A binary mask is created by selecting a specified color and marking the corresponding pixels as white, referred to as foreground, and everything else as black pixels, referred to as background. To improve robustness we can introduce color threshold specifying how much can pixel color vary from the specified color so it is still marked as foreground.

2.4 Euclidean Distance Transform

Euclidean distance transform, referred to as EDT, of binary image assigns to each pixel the Euclidean distance to the nearest specified pixel. This can be used for transforming widths and distances from image to real-world units.

In our application, this algorithm is used to extract the widths of roads or rivers from the segmentation image and convert them to approximate real-world widths. Another use case is the construction of a signed distance field, or SDF.

An SDF is a field in which the value corresponding to each pixel represents the distance from that pixel to the nearest edge of a foreground region. The value is negative if the pixel belongs to the background and positive if it belongs to the foreground.

2.5 Binary Mask Smoothing

Skeletonization algorithms, described below, are sensitive to binary mask "corners". This behavior is most visible by performing skeletonization of wider foreground regions. To minimize this behavior we smooth the edges of binary mask.

Binary mask smoothing is achieved by using Gaussian blur on computed signed distance field, then converting pixels of specified value to foreground pixels and the rest as background pixels. If the algorithm is used for smoothing thinner foreground regions it can result in disrupting their continuity.

User can modify Gaussian blur parameter, σ, to increase or decrease the blur strength, or value threshold to modify what smoothed SDF values will be considered as foreground pixels. If the threshold is lower than zero, pixels will be added to the mask, and if its greater than zero, the pixels will erode.

2.6 Binary Mask Skeletonization

Thinning of a binary mask, referred to as skeletonization, is a process of removing boundary pixels of a foreground region while maintaining their basic structure, general shape and connectivity. For algorithms used to create binary mask skeleton used for this application, we need to ensure satisfaction of certain criteria:

- The resulting line should be 1 pixel wide with no redundant pixels
- Connectivity of individual components must be preserved
- The skeleton should be approximately centered with respect to the local thickness of the component.
- There should not be substantial gap between skeleton and edge of binary mask, if the foreground region is touching the side.

Many thinning algorithms have been proposed for various applications, one of the popular ones are Guo-Hall algorithm and Zhang-Suen algorithm. Their main advantage is their ability to be parallelized, but both of them do not guarantee desired 1-pixel width without some additional post-processing. A different algorithm was selected, which is sequential and guarantees skeleton with no redundant pixels and 1-pixel width.

This algorithm works by performing passes in all four directions and sequentially removes redundant pixels until only the skeleton remains. This results in the skeleton being offset by 1 pixel from center, but this error is insignificant and does not negatively affect the final result.

Additional advantage to its 1-pixel width, is that if foreground region of binary mask is touching image boundary, the algorithm will not create substantial gap between first point of the skeleton and side of the image. For example Zhang-Suen algorithm does not have this feature and it can result in unrealistic rivers or roads.

2.7 Binary Mask Contours

Extracting contours of binary mask foreground region is used for creating features that fill area, such as biomes or lakes.

Used algorithm cycles through every pixel of binary mask. If the pixel is black, that means in background, we skip it. If the pixel is white, in foreground, we count how many 4-neighbors are black and if there are any, we mark this pixel as contour. Pixels that are 4-neighbors are the pixels directly above, below, left or right of the initial pixel.

This algorithm skips foreground pixels along the edges of binary mask, as there are no background pixels. To create edge contour we count pixels that are outside the binary mask bounds as background.

This algorithm, although simple, reliably detects contours. There are a few scenarios where strictly 1-pixel width of the contours is not ensured. These scenarios are hard to resolve by algorithm as their outcome can rapidly change the shape or size of the contours. We decided to leave the resolution of these situations to user by giving them tools for manual edit.

3 Curve Representation of Extracted Topology

Curve representation of extracted topology is done in three main steps. First we convert pixels of the topology to polylines, then we reduce redundant pixels and then we convert polylines to curves.

3.1 Problem Formulation

3.2 Preliminaries

3.3 Endpoint and Junction Pixels Detection

One of the important parts of creating polylines from topological information is detecting which pixel is endpoint and which is junction. We use simple detection by counting how many 8-neighbor pixels of the initial pixels are foreground, or white.

If the resulting amount is one, then we know the initial pixel is endpoint, has only one neighbor. If there are two foreground pixels, then the initial one is normal pixel connecting two other.

Junction has three or more foreground pixels, but generally only three. If there are four or more, it could for example mean there is a plateau or other shape that could potentially make polyline extraction difficult.

3.4 Polyline Creation Algorithm

Polyline creation algorithm uses topological information such as binary mask skeleton or contour to create continuous polyline containing all pixels present in the topology. This algorithm uses information about endpoint and junction pixels, and works by going through three steps in order.

Polyline from Endpoint to Junction or Endpoint

First set is to create polyline from endpoint to junction or another endpoint. We keep track of visited pixels and all endpoints and junction. We start by adding starting endpoint to our polyline and marking it as visited. Then we search for foreground pixels in 8-neighbor pixels.

If foreground pixel is found, if it is already visited, we skip it. Then we check if it is another endpoint, junction or neither. If the pixel is endpoint, we mark it as visited, add it to our polyline and return the resulting polyline. Similarly if it is junction, we add it to our polyline and return the resutl, but we do not mark it as visited. Lastly, if the pixel is neither, we add it to our polyline, mark it as visited and search for foreground pixels around it and we repeat this step.

To increase accuracy and continuity of resulting polyline, we can chose junction or endpoint pixel depending on which is closer to the initial one. We used Manhattan distance for its low computing cost. If selection algorithm is not used, it can result in discontinuous polyline or small redundant segments.

Algorithm 1 Polyline from Endpoint to Junction or Endpoint algorithm

```
procedure ENDPOINT_TO_ENDPOINT(starting_endpoint, visited)
    queue ← initialize queue
    polyline ← initialize empty polyline

    queue.add(starting_endpoint)
    while queue do
        point ← queue.pop()

        visited[ point ] = true
        polyline.add(point)

        best_junction_candinate ←  from neighbors of point
        if best_junction_candinate exists then
            polyline.add(best_junction_candinate)
            return polyline
        end if

        for neighbor from neighbors of point do
            if neighbor is foreground and not visited[ neighbor ] then
                queue.add(neighbor)
                break
            end if
        end for
    end while

    return polyline
end procedure
```

This algorithm is applied to all starting endpoints that are not yet visited.

- **Polyline from Junction to Junction**
- **Polyline from Closed Contour**

4 How to write thesis in LaTeX

4.1 Versioning with git

Write the LaTeX in such a way that it could be versioned by git, which will help when collaborating with other people. This means writing **one sentence per line**. Even when you use third-party platforms, such as the OverLeaf, you can still share the repository through Git.

4.2 Forming paragraphs

A paragraph is formed in LaTeX by an uninterrupted block of non-empty lines. It is recommended to keep a single sentence per line (helps with versioning using git). A new paragraph is started after an empty line.

This is a new paragraph. It is strongly recommended to **avoid** the use of the *newline* (\\) feature of LaTeX for forming paragraphs as it doesn't format the new paragraph properly (no space at beginning of the new paragraph).

4.3 Linguistic anti patterns

Narrative

We recommend to write your thesis in plural form of the first-person narrative in combination with passive tense, e.g.:

- We discourage the use of any other form, and/or
- any other form is discouraged, but **not**
- I discourage you from using the first-person narrative.

Moreover, avoid "instructional" or "teacher"-like style of writing, such as "Now, we multiply the matrix $\mathbf{A}$ by the scalar c to get the scaled matrix $\mathbf{B}$." A better way of writing the same information would be e.g. "Now, the scaled matrix $\mathbf{B}$ is obtained by multiplying the matrix $\mathbf{A}$ by the scalar c."

Pronouns

The use of pronouns (it, this, they) is strongly **discouraged**. Although, pronouns make it easier for you as a writer to form the flow of the text, pronouns also make it much more difficult for the reader to follow the text. The reader is forced to retain more of the context to substitute and understand what the author meant. Moreover, pronouns can easily become vague (there is more than one way how to interpret them) and can become invalid while making editorial changes to the text, i.e., when moving sentences around. A technical text should be written in a way that makes it as easy to read and comprehend as possible and as hard to misunderstand or misinterpret as possible at the same time.

4.4 Mathematical notation with LaTeX

Take care to use the correct mathematical symbols and common ways of denoting mathematical concepts. Use bold fonts to visually distinguish vectors and matrices ($\mathbf{x}$, $\mathbf{A}$) and

scalars (k, N).

Common errors

A frequent error, carried over from programming languages, is using the asterisk symbol ($*$) to denote multiplication. The asterisk correctly denotes convolution. Similarly, the cross sign ($\times$) typically denotes the cross product (it can also used for stating dimensions, such as $10\,\mathrm{m} \times 10\,\mathrm{m}$) and thus should not be used for scalar multiplication. In English mathematical notation, **scalar multiplication is typically not denoted at all**.

This custom may sometimes make it unclear whether a sequence of letters denotes multiplication of several scalars or a multi-letter variable, such as

$$T = T0 + coeffmeas, \tag{4.1}$$

where the variables in this hypothetical equation are T, $T0$, $coeff$ and $meas$. For this reason, **avoid using multi-letter variable naming** and strive to denote mathematical variables with single letters optionally a with lower or upper index, or other modifiers (`\hat`, `\bar`, etc.). The equation above could be modified to be

$$T = T_0 + cT_{\mathrm{meas}}. \tag{4.2}$$

If the multiplication is still unclear (e.g. when multiplying many single-letter scalars), the `\cdot` symbol may be used such as

$$P \cdot V = n \cdot R \cdot T. \tag{4.3}$$

Equations

Mathematical equations should be numbered and should be a part of a sentence. For example, a discrete LTI system update is described as

$$\mathbf{x}_{[k+1]} = \mathbf{A}\mathbf{x}_{[k]} + \mathbf{B}\mathbf{u}_{[k]}, \tag{4.4}$$

where $\mathbf{x}_{[k]} \in \mathbb{R}^m$ is the state vector at the sample k, $\mathbf{u}_{[k]} \in \mathbb{R}^n$ is the input vector, $\mathbf{A} \in \mathbb{R}^{m\times m}$ is the main system matrix, and $\mathbf{B} \in \mathbb{R}^{m\times n}$ is the system input matrix. Proper punctuation should be used after the equation, as if it were an ordinary object in the sentence.

Do not put any empty lines before the equation. If the sentence that the equation is a part of continues after the equation (as is the case here), do not put empty lines after the equation either. That would create a new paragraph mid-sentence. For an example of how not to do it, the equation

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{4.5}$$

describes the logistic function often used in machine learning. Observe how a new paragraph is created for the equation and then for this block of text (compare with the proper typesetting above). Not only does this not look correct, it may also cause incorrect page breaking.

4.5 Using footnotes

Do not be afraid to use footnotes for additional information, such as http links[1]. We use footnote links whenever we want to *point* to a website, rather then to cite it as a source. Like with everything, do not overdo it.

4.6 Referencing document elements

LaTeX allows you to dynamically reference to parts of the documents, such as

- figures: Fig. 4.4, Figure 4.4,
- equations: eq. (4.4), (4.4),
- code: Lst. 4.1,
- and any other object that can contain a `\label`.

Check the section in the `document_setup.tex` that contains useful macros for unifying the references:

```
\newcommand{\reffig}[1]{Fig.~\ref{#1}}
\newcommand{\reflst}[1]{Lst.~\ref{#1}}
\newcommand{\refalg}[1]{Alg.~\ref{#1}}
\newcommand{\refsec}[1]{Sec.~\ref{#1}}
\newcommand{\reftab}[1]{Table~\ref{#1}}
\newcommand{\refeq}[1]{\eqref{#1}}
```

Listing 4.1: LaTeX macros for referencing to document elements.

4.7 Abbreviations with Acronym

Abbreviations are handled by the *acronym* package. Example sentence with abbreviations: "UAV is a flying vehicle that commonly uses Light Detection and Ranging (LiDAR) and Global Positioning System (GPS) receiver". Note that the acronyms are only explained once in the document by default. It is good practice to re-explain acronyms used both in the abstract and the rest of the document as the abstract is often presented separately. This can be achieved by resetting the internal status of the acronyms ("forgetting" that they were explained) using the `\acresetall` command after the abstract. Please, read the documentation[2].

4.8 Units of measurements with Siunitx

Typesetting of units has never been more accessible with the Siunitx package. Acceleration is measured in $\mathrm{m\,s^{-2}}$. Gravity accelerates objects at a rate $\approx 9.81\,\mathrm{m\,s^{-2}}$ near the sea level. You can define your units if you want.

4.9 Hyphens and dashes

Hyphens and dashes are the various form of the symbol "-" used in many situations. There are also various ways how to typeset the symbol in LaTeX.

- The *hyphen* is used to compound words, e.g., "the eye-opener". The hyphen is typeset as a single *minus/hyphen* character: `-`.

[1] This repository: https://github.com/ctu-mrs/thesis_template.
[2] Acronym package: http://mirrors.ctan.org/macros/latex/contrib/acronym/acronym.pdf

- The *en-dash* is used to specify ranges of values, e.g., "between 2–10". The en-dash is typeset as two consecutive hyphens characters: `--`.
- The *em-dash* is used to separate complex sentences in place of commas, parenthesis and colons — each with its particular rules. The em-dash is typeset as three consecutive hyphens characters: `---`.

Check the https://www.thepunctuationguide.com/ for all the details.

4.10 Double quotation marks

"Double quotes" in English are composed of a pair of opening (") and closing (") symbols. The opening symbol is typeset as two backtick characters: `` ``` `` (typically below the `Esc` key on the English keyboard), and the closing quotes as two apostrophes: `''`. The LaTeX engine will convert them automatically to the opening and closing symbols. A more robust solution is to use the `csquotes` package and the `\enquote` command which also takes care of nested quoting and other peculiarities.

4.11 2D Diagrams with Tikz

Tikz is a powerful tool for drawing 2D (and 3D) shapes and diagrams. Check the documentation and examples: https://www.overleaf.com/learn/latex/TikZ_package. The benefit of using *Tikz*, instead of some other third-party drawing program, are:

- fonts are the same as in LaTeX,
- you can typeset math in LaTeX,
- you can use references to other parts of your document,
- you can version the image in git,
- the images are easily adjustable while editing your document.

Check Fig. 4.1 for example.

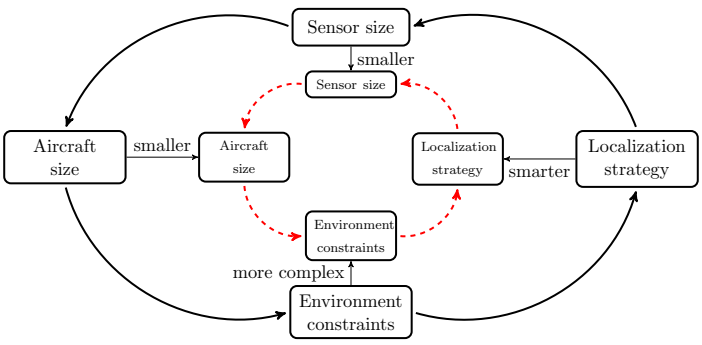

Figure 4.1: Example of a 2D diagram using tikz *PGFPlots*.

4.12 Data plots with PGFPlots

PGFPlots produces nice 2D and 3D data plots from data stored in CSV. The plot parameters can be versioned and easily adjusted by editing the plot definition file.

- Documentation and manual: https://ctan.org/pkg/pgfplots
- Compile the plots individually and then include the pdfs because it can take longer.
- Example located in `fig/plots/example_plot`, see Fig. 4.2.

- You could include the latex file directly. However, it will take longer to compile, and platforms such as Overleaf can have a problem with that.

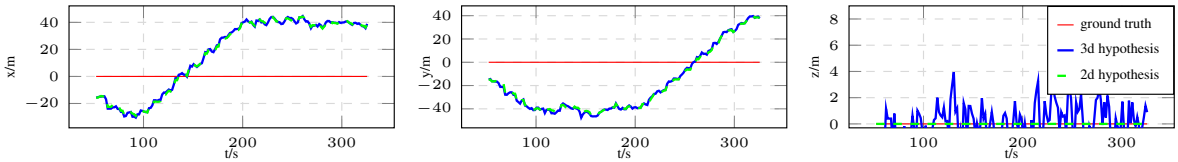

Figure 4.2: Example of a 2D plot using *PGFPlots*.

4.13 3D Plots with Sketch

Sketch is a tool for defining a 3D scene using simple descriptive language. The 3D scene is then converted to *Tikz*, which is later compiled to pdf. The benefits of using *Sketch* are similar to using *Tikz*: LaTeX fonts, versioning using git, and cleanness of the result. See the example image in Fig. 4.3.

- Documentation and manual: http://www.frontiernet.net/~eugene.ressler/
- Cross-compilation from *Sketch* to *pdf* using the `fig/sketch/compile_sketch.sh` script.

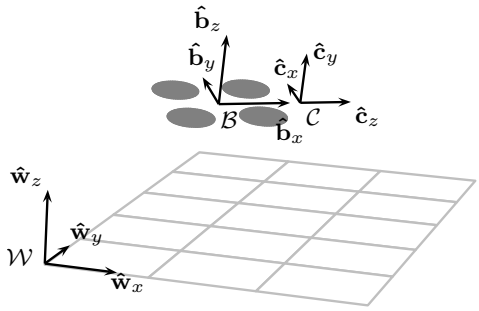

Figure 4.3: Depiction of the used coordinate systems. The image was drawn using *Sketch*.

4.14 Image collages with Subfig

We recommend using the `subfig` package, which provides the `\subfloat` command. It is more versatile than the simpler `subcaption` package. Check Fig. 4.4 for an example.

4.15 Citations with Biblatex

Biblatex is probably the most powerful citation package for LaTeX. It consumes the standard `.bib` file. However, it can sort and filter the citations using the `keywords` tag. Citing references is done using the `cite` command, e.g., [1]. You can also define some nice citation boxes, such as this one:

[1] T. Baca, M. Petrlik, M. Vrba, V. Spurny, R. Penicka, D. Hert, et al., "The MRS UAV System: Pushing the Frontiers of Reproducible Research, Real-world Deployment, and Education with Autonomous Unmanned Aerial Vehicles," *Journal of Intelligent & Robotic Systems*, vol. 102, no. 26, pp. 1–28, 1 May 2021

(a) A UAV, the T650 model.

(b) Another UAV, again, the T650 model.

Figure 4.4: The caption should mention both subfigures, the Fig. 4.4a and the Fig. 4.4b. You can just refer to them as (a) and (b) in the main Figure's caption, but beware, you need to keep it correct as you edit.

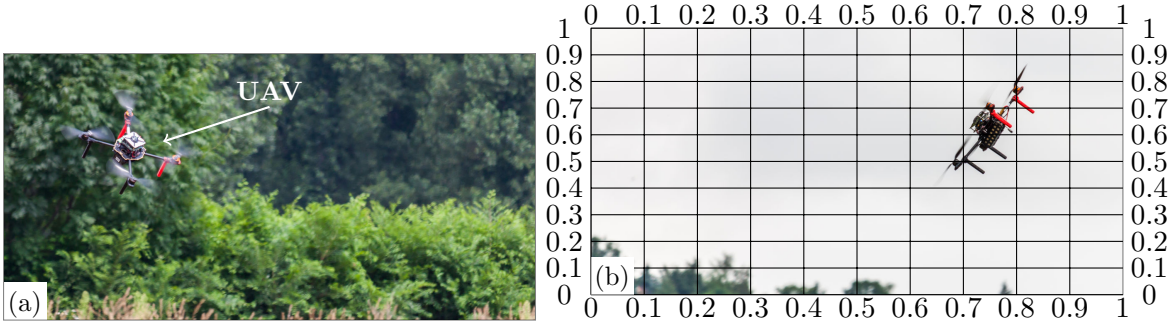

Figure 4.5: Example of using Tikz for image overlays. (a) shows a final product, (b) shows a grid useful for nailing down the coordinates.

4.16 Image overlays with Tikz

Tikz is very useful to create custom image overlays. The overlay can be set such that the image is spanned by Cartesian coordinates $(x, y) \in [0, 1]^2$ Example can be seen in Fig. 4.5.

4.17 General tips

In general, strive to make the paper easy to read and understand, and hard to misunderstand or misinterpret. Here are some more specific tips on how to achieve that (and other general suggestions).

- **Be consistent.** This applies in all contexts. For example, if you decide to use the name "LiDAR", do not mix it with "LIDAR" or "Lidar", do not mix different mathematical notations, ensure your Figures have the same style and use the same graphics for the same concepts, etc.
- After you finish writing or modifying any of:
  - a sentence,
  - a paragraph,
  - a section/chapter,
  - the whole paper/thesis,

  re-read it to make sure that it makes sense, it is coherent and correct, and doesn't

contain typos.

- If you're using a LLM-based tool (ChatGPT etc.) for grammar-proofing or even formulation of sentences, **do not just copy-paste its response** to your query. The previous rule applies doubly here. LLMs tend to often produce confident-sounding nonsense, sentences with reformulated duplicated content, or with a slightly changed meaning. They are a good tool to get inspiration to start writing about a subject, for grammar-checking, or for finding alternative, nice-sounding formulations, but they can lie or warp facts — take care when using them!

5 Conclusion

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

6 References

[1] T. Baca et al., "The MRS UAV System: Pushing the Frontiers of Reproducible Research, Real-world Deployment, and Education with Autonomous Unmanned Aerial Vehicles," *Journal of Intelligent & Robotic Systems*, vol. 102, no. 26, pp. 1–28, 1 May 2021.

[2] T. Baca, G. Loianno, and M. Saska, "Embedded Model Predictive Control of Unmanned Micro Aerial Vehicles," in *IEEE International Conference on Methods and Models in Automation and Robotics (MMAR)*, IEEE, 2016, pp. 992–997.

[3] A. Benallegue, A. Mokhtari, and L. Fridman, "High-order sliding-mode observer for a quadrotor UAV," *International Journal of Robust and Nonlinear Control: IFAC-Affiliated Journal*, vol. 18, no. 4-5, pp. 427–440, 2008.

A Appendix A